

Develop a **Flutter-based Smart Study Planner & Exam Preparation Tracker App** that helps students plan subjects, schedule study sessions, track syllabus completion, and analyze preparation progress with offline-first functionality and synchronized updates.

Students preparing for exams often face difficulties such as:

- Poor time management and unstructured study plans
- Difficulty tracking syllabus completion
- Lack of visibility into preparation progress
- Overlooking important subjects or topics
- Dependency on internet-based tools

To address these challenges, a mobile application is required that enables students to **plan study schedules, track completion status, and analyze preparation progress effectively**, even in offline scenarios.

Design and implement a mobile application that:

- Allows students to plan study schedules
- Tracks subject-wise and topic-wise progress
- Provides insights into preparation status
- Helps in prioritizing subjects based on completion
- Supports offline usage with later synchronization

Functional Requirements

1. Subject & Topic Management Module

- Add subjects with:
 - Subject Name
- Add topics under each subject:
 - Topic Name
 - Estimated study time

2. Study Scheduling Module

- Schedule study sessions with:
 - Subject
 - Topic
 - Date & Time
 - Duration

3. Progress Tracking Module

- Mark topics as:
 - Not Started
 - In Progress
 - Completed
- Track percentage completion per subject

4. Study Dashboard

- Display:
 - Total subjects
 - Completed topics
 - Pending topics
 - Daily study progress
- Visual indicators (progress bars/charts)

5. Priority & Planning Logic

- Highlight:
 - Subjects with lowest completion
- Suggest next topics to study (basic logic)

6. Search & Filter Module

- Search topics by name
- Filter by:
 - Subject
 - Status (Completed / Pending)
 - Date

7. Offline Functionality

- Allow scheduling and tracking without internet
- Store data locally
- Sync updates when online

8. Validation & Error Handling

- Prevent invalid schedules (past dates optional check)
- Validate required fields
- Provide meaningful alerts

Technical Requirements

- Framework: **Flutter**
- State Management: Provider / Riverpod / Bloc
- Local Storage: Hive / SQLite
- Optional Backend: Firebase / REST API
- Modular architecture (Subject, Schedule, Progress modules)
- Clean and maintainable code

UI/UX Requirements

Design a minimum of **five screens**:

1. Subject & Topic Management Screen
 2. Study Scheduling Screen
 3. Study Progress Screen
 4. Dashboard Screen
 5. Search & Filter Screen
- Use visual progress indicators
 - Ensure easy navigation between planning and tracking
 - Maintain clean and structured layout

GitHub Requirement

- Maintain a public GitHub repository
- Minimum **4 commits**:
 1. Project Initialization
 2. UI Implementation
 3. Core Logic (Scheduling + Progress Tracking)
 4. Offline Storage & Final Enhancements

Submission Requirements

PDF Name: StudentID_StudyPlannerApp.pdf

Must Include:

- Screenshots of all UI screens
- Description of implemented modules
- Explanation of progress tracking and priority logic
- Future Scope
- Conclusion